Main-Quantitative_Data

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```
library(tinytex)
library(ggplot2)
```

1. Organise and clean the data

1.1 Subset the data into the specific dataset allocated

```
SID <- 
SIDoffset <- (SID %% 50) + 1
load("carbon-footprint.Rda")
mydf <- df[seq(from=SIDoffset,to=nrow(df),by=50),]
```

1.2 Data quality plan

I began by examining the overall structure of the dataset to verify how R had interpreted each variable and to understand the contents of my subset, which consisted of 240 rows and 11 variables. I checked the first and last six rows to ensure the data were being read correctly by R. Then identified the numerical variables (household_size, annual_energy_consumption, carbon_footprint, environmental_concern_score, income, number_adults) and the categorical variables (recycles_regularly, uses_public_transport, diet_type, home_type, recycles_regularly_question). After eyeballing the dataset, I checked all variables for missing values and found that income contained 22 missing observations. I then compared each variable against the metadata to identify unexpected values, which revealed that annual_energy_consumption included an implausible minimum of –650, and in total, four values fell below the expected lower limit of 1000. I also reviewed the categorical variables for misspellings or duplicated categories and observed inconsistencies in home_type. And finally, I checked the relationship between household_size and number_adults. I found 28 rows where number_adults was zero, and household_size was 1, indicating households with no adults but one child, which seemed unusual. Excluding these 28 rows, both variables aligned perfectly across the remaining data.

1.3 Data quality findings

Issue 1 - Implausible values in annual_energy_consumption

After careful observation of the summary output and comparing the values with the metadata, I identified that the variable annual_energy_consumption contains implausible values. The expected range for this variable is between 1000 and 9000, but the data includes some values falling below this threshold. In total, four rows contained values

under 1000, including one extreme value of –650, which is impossible for annual energy consumption. These values suggest either data entry errors or incorrect measurements and therefore require cleaning before further analysis.

```
mydf[mydf$annual_energy_consumption < 1000 | mydf$annual_energy_consumption > 9000, ]
```

```
##      household_size annual_energy_consumption recycles_regularly
## 3043              1                     -500                   1
## 3093              1                     -500                   0
## 6043              6                     -650                   1
## 6093              2                     -650                   0
##      uses_public_transport diet_type carbon_footprint
## 3043                   Yes  Omnivore          10814.4
## 3093                    No     Vegan           7170.5
## 6043                    No  Omnivore          13117.7
## 6093                    No  Omnivore           9862.6
##      environmental_concern_score   income number_adults home_type
## 3043                           8       NA             1      Flat
## 3093                           4       NA             1  Detached
## 6043                           4 22353.31             6  Detached
## 6093                           3 12987.12             2      Flat
##      recycles_regularly_question
## 3043                         Yes
## 3093                          No
## 6043                         Yes
## 6093                          No
```

Issue 2 - Missing values in income

While checking for missing entries, I discovered that the variable “income” contained some missing values. This was identified after examining each variable for NA entries. In total, 22 rows had missing income data. Since income is a key demographic variable and will likely be used in future modelling, these missing values represent a clear data quality concern. They could impact the reliability of any analysis involving income. No other numerical variable exhibited similar levels of missingness.

```
mydf[is.na(mydf$income), ]
```

```
##      household_size annual_energy_consumption recycles_regularly
## 2043              3                    6064.4                   1
## 2093              3                    3865.5                   0
## 2143              1                    3019.1                   1
## 2193              3                    5476.6                   0
## 2243              4                    4088.9                   0
## 2293              5                    4405.7                   0
## 2343              3                    4090.0                   1
## 2393              3                    5125.1                   0
## 2443              3                    5060.3                   0
## 2493              4                    4890.5                   1
```

```
## 2543              1                    3673.4                   0
## 2593              3                    5532.5                   1
## 2643              3                    5269.7                   0
## 2693              1                    4865.0                   0
## 2743              2                    3967.2                   1
## 2793              2                    4716.7                   1
## 2843              3                    4443.7                   1
## 2893              2                    4856.1                   0
## 2943              1                    3904.5                   1
## 2993              2                    2268.9                   1
## 3043              1                    -500.0                   1
## 3093              1                    -500.0                   0
##      uses_public_transport   diet_type carbon_footprint
## 2043                   Yes    Omnivore          13699.0
## 2093                    No Pescatarian          11062.6
## 2143                    No    Omnivore           7630.6
## 2193                    No  Vegetarian          15000.0
## 2243                   Yes    Omnivore           9862.2
## 2293                    No    Omnivore          14325.1
## 2343                   Yes    Omnivore           9232.5
## 2393                    No    Omnivore          12288.1
## 2443                    No    Omnivore          14900.5
## 2493                   Yes Pescatarian          11658.7
## 2543                    No    Omnivore           9659.8
## 2593                   Yes    Omnivore          15000.0
## 2643                    No    Omnivore          14561.6
## 2693                    No Pescatarian          11958.5
## 2743                   Yes  Vegetarian           9346.1
## 2793                    No  Vegetarian          10951.6
## 2843                    No    Omnivore          10899.8
## 2893                    No       Vegan           9895.2
## 2943                   Yes    Omnivore           9185.3
## 2993                   Yes    Omnivore           5892.4
## 3043                   Yes    Omnivore          10814.4
## 3093                    No       Vegan           7170.5
##      environmental_concern_score income number_adults     home_type
## 2043                           9     NA              3      Terrrace
## 2093                           4     NA              3      Terrrace
## 2143                           6     NA              1 Semi-detached
## 2193                           4     NA              3      Detached
## 2243                           6     NA              4          Flat
## 2293                           5     NA              5          Flat
## 2343                           7     NA              3          Flat
## 2393                           5     NA              3      Terraced
## 2443                           3     NA              3      Terraced
## 2493                           7     NA              4          Flat
## 2543                           2     NA              0      Detached
## 2593                           6     NA              3      Terraced
## 2643                           4     NA              3      Terraced
## 2693                           4     NA              1      Terraced
```

```
## 2743                            7     NA              2      Terraced
## 2793                            5     NA              2          Flat
## 2843                            5     NA              3          Flat
## 2893                            6     NA              2      Detached
## 2943                            7     NA              1      Terraced
## 2993                            6     NA              2      Terraced
## 3043                            8     NA              1          Flat
## 3093                            4     NA              1      Detached
##      recycles_regularly_question
## 2043                         Yes
## 2093                          No
## 2143                         Yes
## 2193                          No
## 2243                          No
## 2293                          No
## 2343                         Yes
## 2393                          No
## 2443                          No
## 2493                         Yes
## 2543                          No
## 2593                         Yes
## 2643                          No
## 2693                          No
## 2743                         Yes
## 2793                         Yes
## 2843                         Yes
## 2893                          No
## 2943                         Yes
## 2993                         Yes
## 3043                         Yes
## 3093                          No
```

```
sum(is.na(mydf$income))
```

```
## [1] 22
```

Issue 3 - Misspellings and inconsistent labels in home_type

While examining the categorical variables, I identified several inconsistencies in the “home_type” variable. The unique values revealed issues with case sensitivity and obvious misspellings. For example, “Flat” appeared as both “Flat” and “flat”, indicating inconsistency. “Terraced” was correctly listed as “Terraced”, but also as “Terrrace”, which is a misspelling. Similarly, “Semi-detached” appeared as “Semi-detached” and “semi-detached”, while Detached was listed as “detatched”. These inconsistencies lead to duplicate categories that represent the same home_type. According to the metadata, there should only be four valid property types: Flat, Detached, Terraced, and Semi-detached. However, due to misspellings and case variations, the data displays eight unique types.

```r
unique(mydf$home_type)
```

```
## [1] "Detached"      "Semi-detached" "Terraced"      "Flat"
## [5] "flat"          "Terrrace"      "semi-detached" "detatched"
```

```r
table(mydf$home_type)
```

```
##
##      Detached     detatched          flat          Flat semi-detached
##            58             2             2            59             2
## Semi-detached      Terraced      Terrrace
##            52            63             2
```

1.4 Data cleaning

Addressing Issue 1 - Implausible values in annual_energy_consumption

I removed the four rows that contained values below 1000 to maintain the range specified in the metadata (1000 to 9000). Since only four rows were affected, removing them ensured that the remaining energy values adhered to the expected measurement scale without significantly reducing the dataset. Imputation was an option, but removal was more suitable given the nature and extent of the error.

```r
mydf <- mydf[mydf$annual_energy_consumption >= 1000 &
mydf$annual_energy_consumption <= 9000, ]
```

Addressing Issue 2 - Missing values in income

I chose to impute the 22 missing income values using the median income from the dataset. Income is typically right-skewed, and the median offers a more stable estimate than the mean because it is less affected by extreme values. Using the median allowed me to retain all households while maintaining the overall income distribution consistent with the observed data. An alternative approach, removing the 22 rows, would have significantly reduced the dataset and was not suitable given the limited sample size.

```r
median_income <- median(mydf$income, na.rm = TRUE)
mydf$income[is.na(mydf$income)] <- median_income
sum(is.na(mydf$income))
```

```
## [1] 0
```

Addressing Issue 3 - Misspellings and inconsistent labels in home_type

I standardized all misspelled and inconsistent entries to include only the four valid categories specified in the metadata: Flat, Detached, Terraced, and Semi-detached. I didn't explore alternative approaches since these entries were clearly misspellings referring to the same types of homes.

```r
mydf$home_type <- replace(mydf$home_type, mydf$home_type == "flat", "Flat")
mydf$home_type <- replace(mydf$home_type, mydf$home_type == "semi-detached",
```

```
"Semi-detached")
mydf$home_type <- replace(mydf$home_type, mydf$home_type == "detatched",
"Detached")
mydf$home_type <- replace(mydf$home_type, mydf$home_type == "Terrrace",
"Terraced")

unique(mydf$home_type)
```

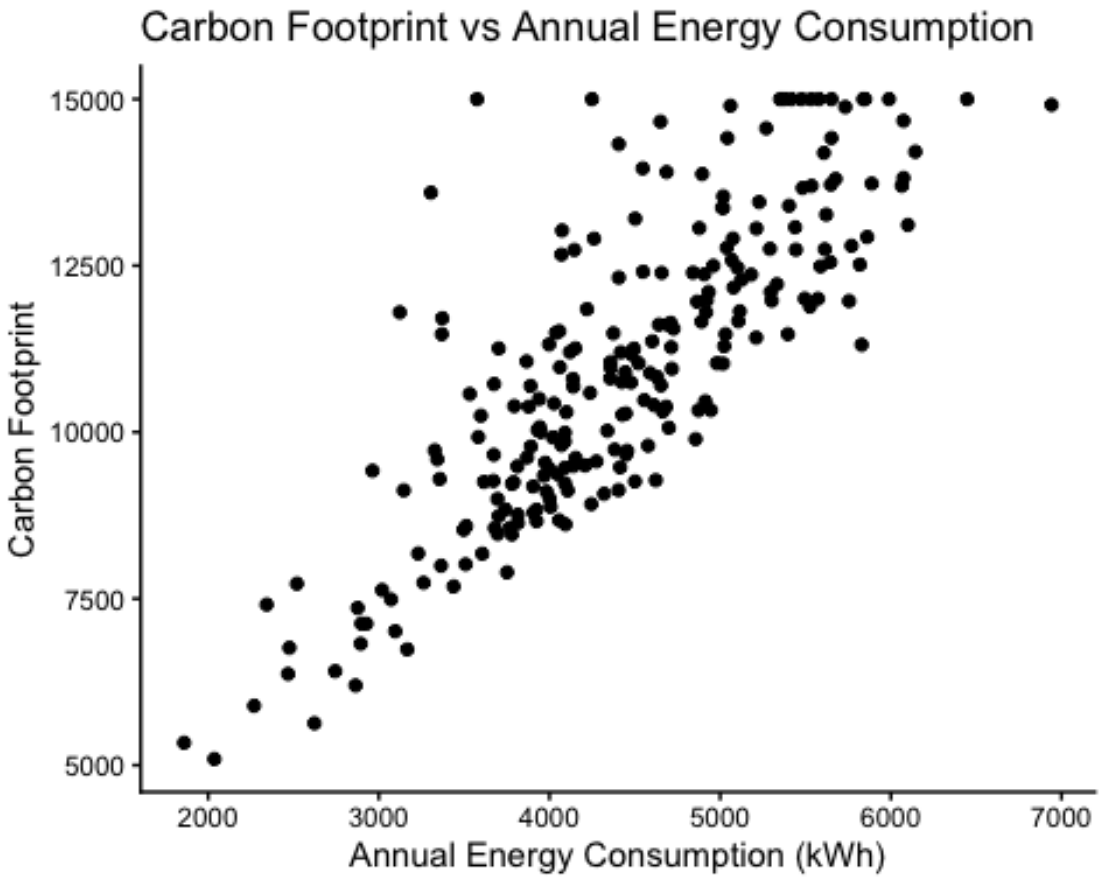
```
## [1] "Detached"      "Semi-detached" "Terraced"      "Flat"
```

2. Exploratory Data Analysis (EDA)

I began by examining each variable individually to understand its distribution and relationship to the dependent variables. I used histograms for numerical variables and bar charts for categorical ones to evaluate their overall spread and frequency patterns. Scatterplots helped identify potential relationships or linear trends between numerical variables and the carbon footprint. I also produced boxplots to compare numerical variables across different levels of the dependent variables, helping to visualise how each predictor behaved. To compare proportions between categorical variables, I used bar charts. Lastly, I reviewed some correlations among numerical variables.

Element 1 - Scatterplot of Annual Energy Consumption vs Carbon Footprint

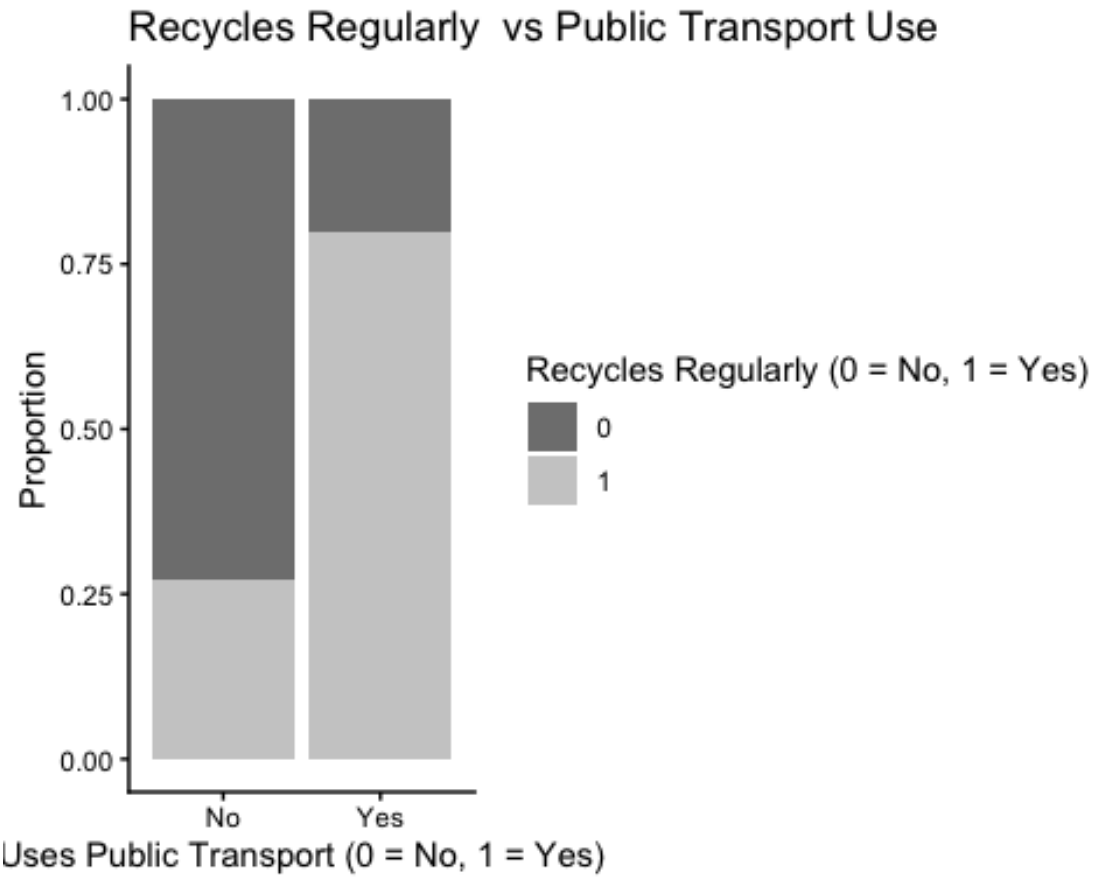
```
ggplot(mydf, aes(x = annual_energy_consumption, y = carbon_footprint)) +
geom_point() + ggtitle("Carbon Footprint vs Annual Energy Consumption") +
xlab("Annual Energy Consumption (kWh)") + ylab("Carbon Footprint") +
theme_classic()
```



The scatterplot reveals a clear positive linear relationship between higher annual energy consumption and consistently elevated carbon footprints. The points form a strong upward trend, suggesting that energy consumption is a key factor influencing emissions. This makes it one of the most valuable variables for modelling carbon_footprint and supports its use as a primary explanatory variable.

Element 2 - Bar Plot of Recycles Regularly vs Public Transport Use

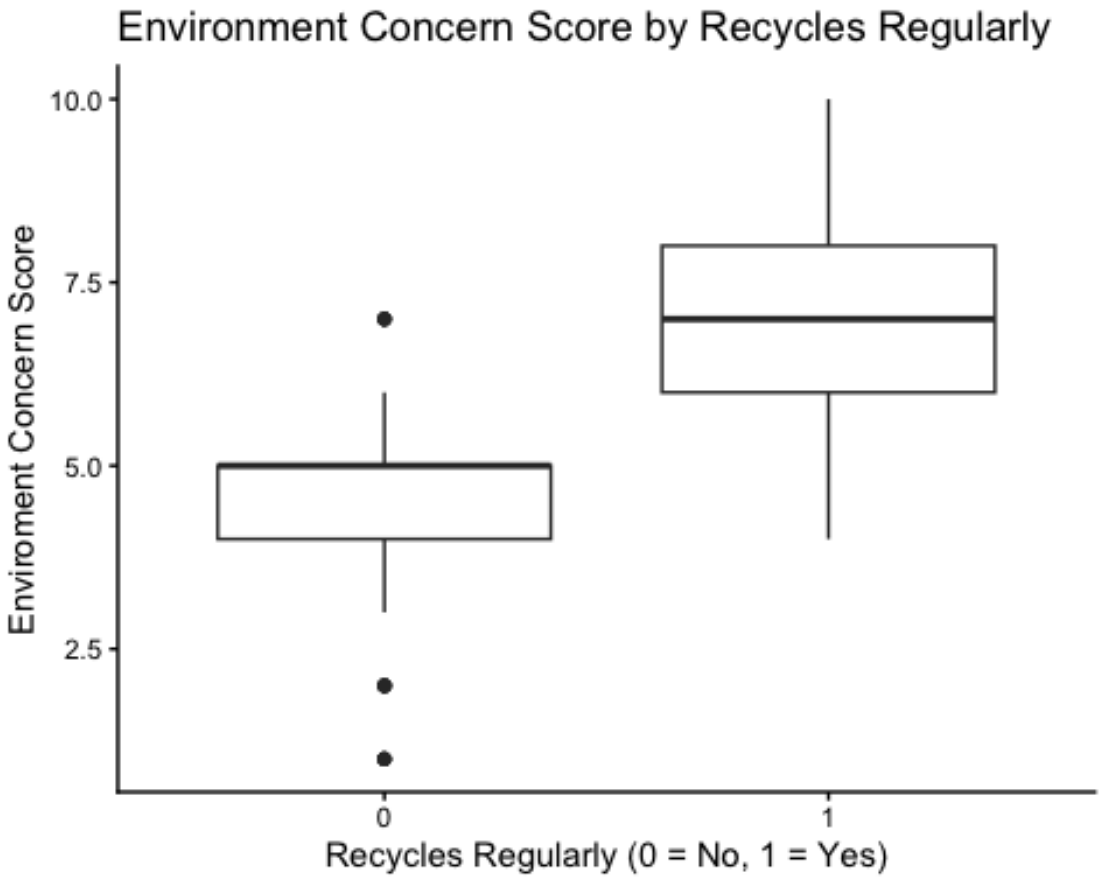
```
ggplot(mydf, aes(x = as.factor(uses_public_transport), fill =
as.factor(recycles_regularly))) + geom_bar(position = "fill") + xlab("Uses
Public Transport (0 = No, 1 = Yes)") + ylab("Proportion") + ggtitle("Recycles
Regularly  vs Public Transport Use") + scale_fill_manual(values = c("grey50",
"grey80"), name = "Recycles Regularly (0 = No, 1 = Yes)") + theme_classic()
```



This bar plot compares the proportion of individuals who recycle regularly among public transport users and non-users. The distribution indicates that those who use public transport have a significantly higher proportion of regular recyclers than non-users. This suggests a possible behavioural link between sustainable transport choices and recycling habits.

Element 3 Boxplot of Recycles Regularly vs Environment Concern Score

```
ggplot(mydf, aes(x = as.factor(recycles_regularly), y =
environmental_concern_score)) + geom_boxplot() + ggtitle("Environment Concern
Score by Recycles Regularly") + xlab("Recycles Regularly (0 = No, 1 = Yes)")
+ ylab("Enviroment Concern Score") + theme_classic()
```



This boxplot indicates that individuals who recycle regularly tend to have higher environmental concern scores. The median concern score is significantly higher for recyclers, with a narrower distribution compared to non-recyclers. This implies a clear behavioural link: greater environmental concern is associated with a higher likelihood of recycling.

3. Modelling

3.1 Explain your analysis

The aim was to model carbon_footprint based on data exploration and observed patterns, using variables that plausibly influence household emissions. I chose a multiple linear regression approach because carbon_footprint is continuous, and early plots indicated approximately linear relationships with some predictors.

My initial model included: annual_energy_consumption, which showed a strong positive linear trend with carbon footprint and was the most influential predictor in the correlations (0.82). Income showed a moderate positive trend with some outliers, suggesting that higher-income households may consume more energy or resources. household_size demonstrated a noisy but generally upward trend. environmental_concern_score displayed slight variation across carbon_footprint. Though weak, its behavioural relevance made it worth testing. Finally, diet_type, uses_public_transport, and home_type showed

differences in carbon_footprint distributions in the boxplots, indicating mild categorical effects.

I excluded recycles_regularly because the EDA showed minimal differences between groups. I also excluded recycles_regularly_question, as both variables represent the same information once converted to binary. Furthermore, I removed number_adults, since household_size already captures the same underlying information (as discussed earlier in the data findings,section 1).

The modelling process began with the selected variables, followed by stepwise AIC selection to identify a more parsimonious specification. Stepwise selection was appropriate because the dataset contains a mixture of continuous and categorical predictors.

The stepwise procedure removed diet_type and environmental_concern_score, which had weak effects, resulting in a simplified model with five predictors: annual_energy_consumption, income, household_size, uses_public_transport, and home_type.

Diagnostic checks for both the initial and selected models showed no strong nonlinearity, but indicated mild curvature at higher fitted values in the residuals-versus-fitted plot. There was slightly heavy-tailed behaviour with a few notable outliers (e.g., 493, 2293) in the Q-Q plot, mild heteroscedasticity in the scale-location plot, and one moderately influential point (493) in the residuals versus leverage plot, not severe enough to invalidate the model.

Weaknesses include slight deviations from normality, mild heteroscedasticity, and influential observations. These were addressed by comparing alternative model specifications through AIC, checking coefficient stability across models, and selecting a simpler model that reduces variance inflation while retaining strong explanatory power (Multiple $R^2 \approx 0.89$).

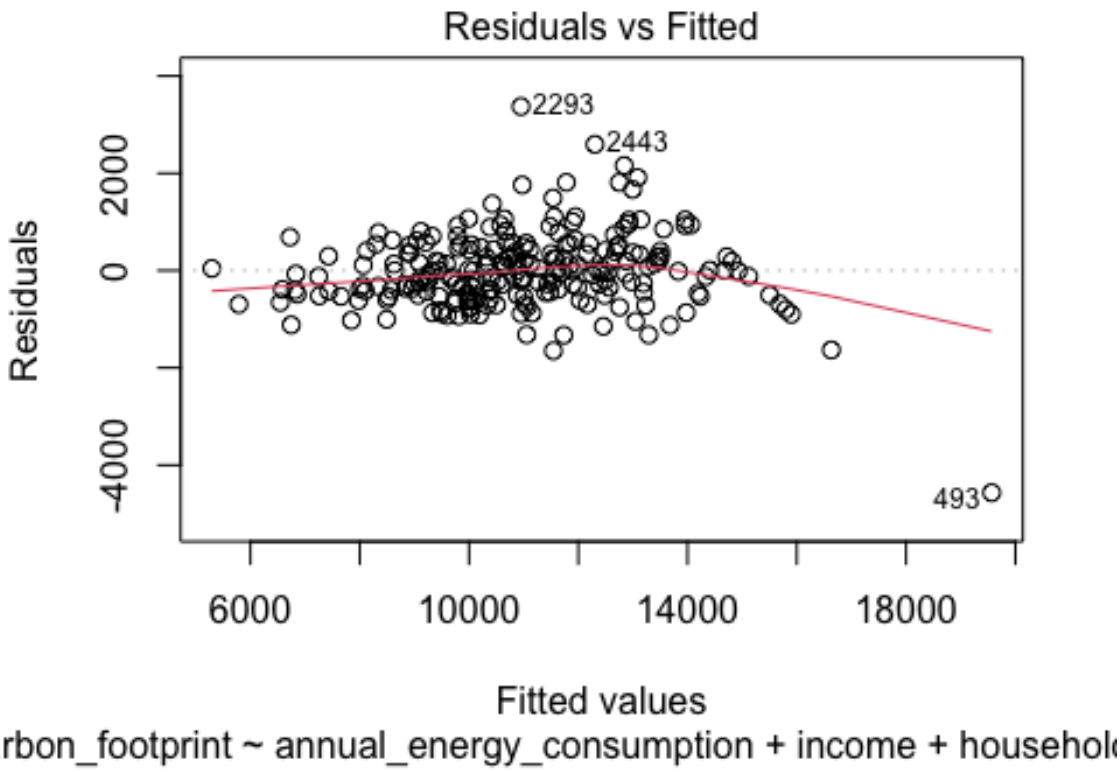
3.2 Provide a model for carbon_footprint.

Initial Model

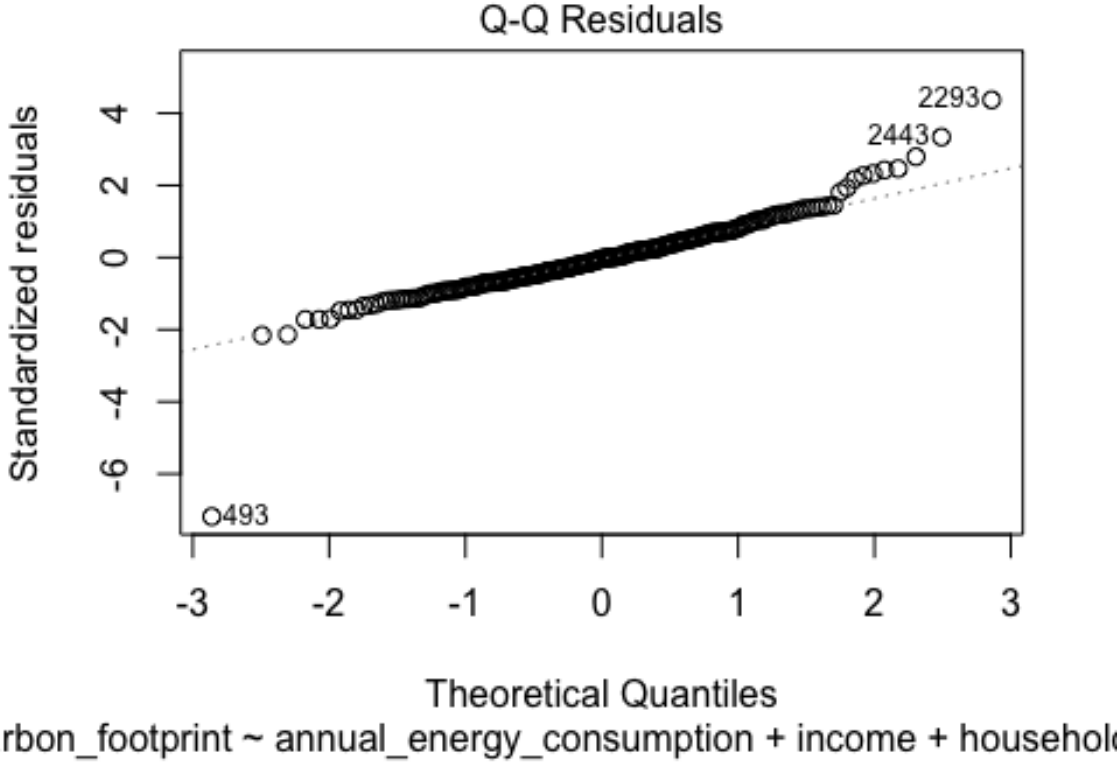
```
initial_model.lm <- lm(carbon_footprint ~ annual_energy_consumption + income
+ household_size + environmental_concern_score + diet_type +
uses_public_transport + home_type, data = mydf)
```

Intial Model diagnostics

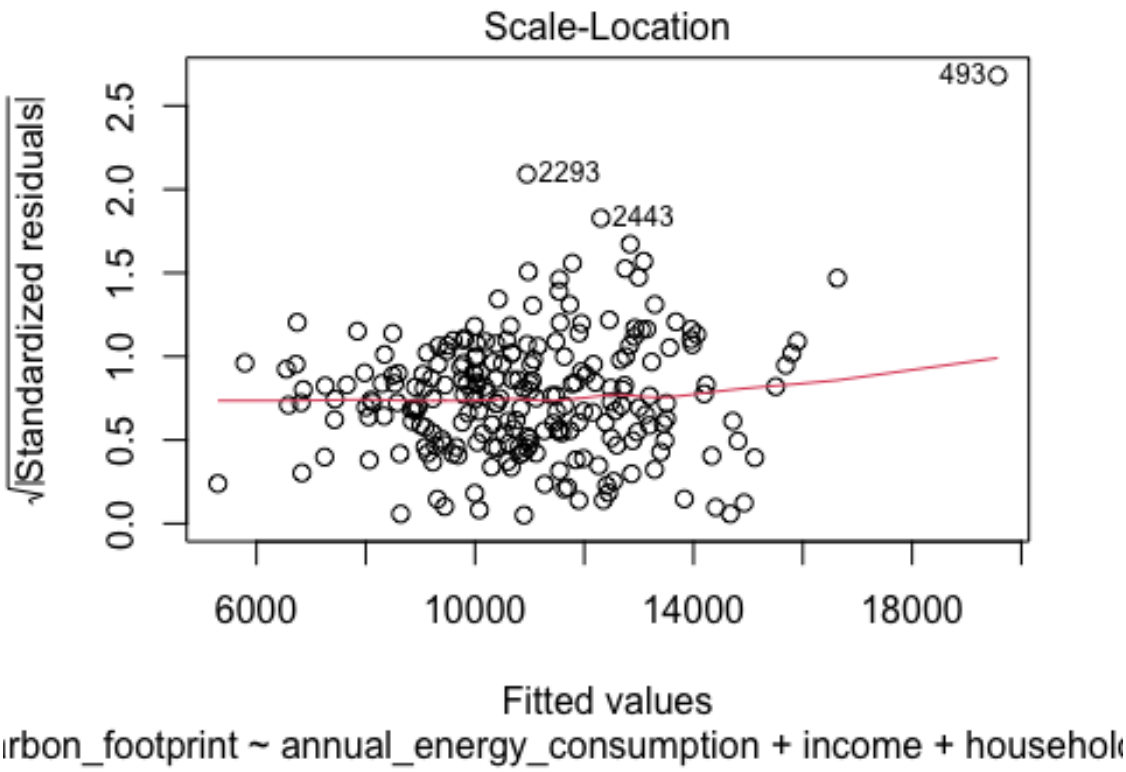
```
plot(initial_model.lm)
```

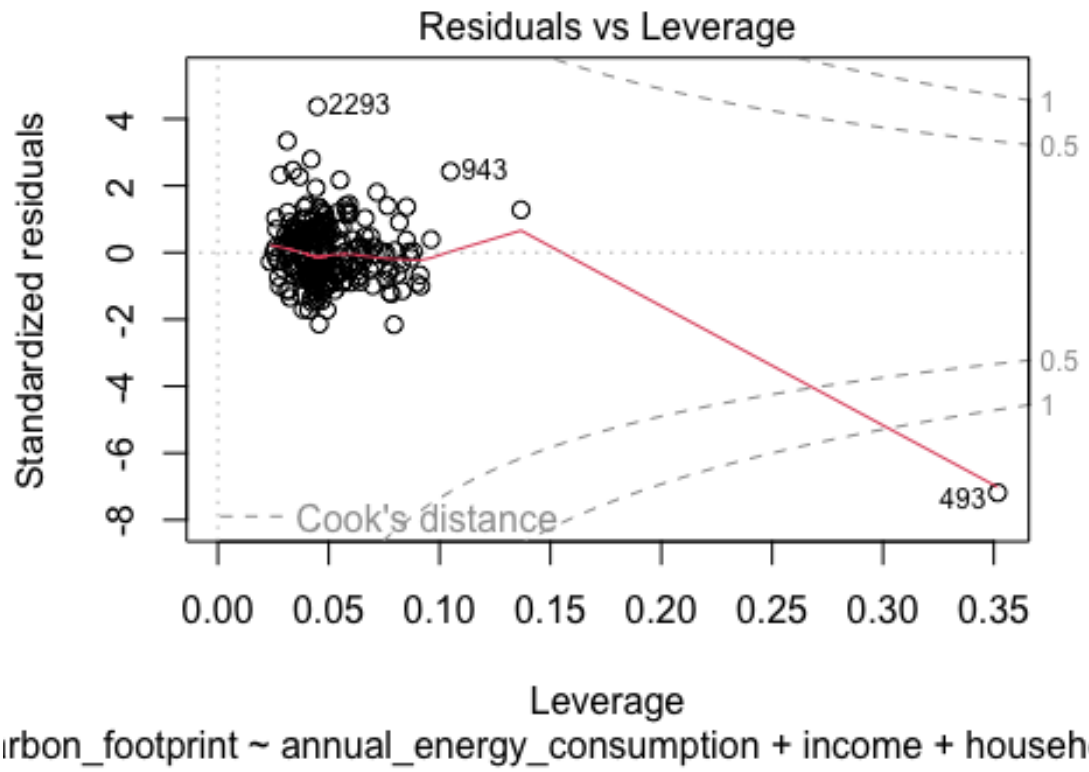


ırbon_footprint ~ annual_energy_consumption + income + household



ırbon_footprint ~ annual_energy_consumption + income + household


Scale-Location
493
2293
2443
√|Standardized residuals|
0.0
0.5
1.0
1.5
2.0
2.5
6000
10000
14000
18000
Fitted values
ırbon_footprint ~ annual_energy_consumption + income + household



rbon_footprint ~ annual_energy_consumption + income + household

Proposed model

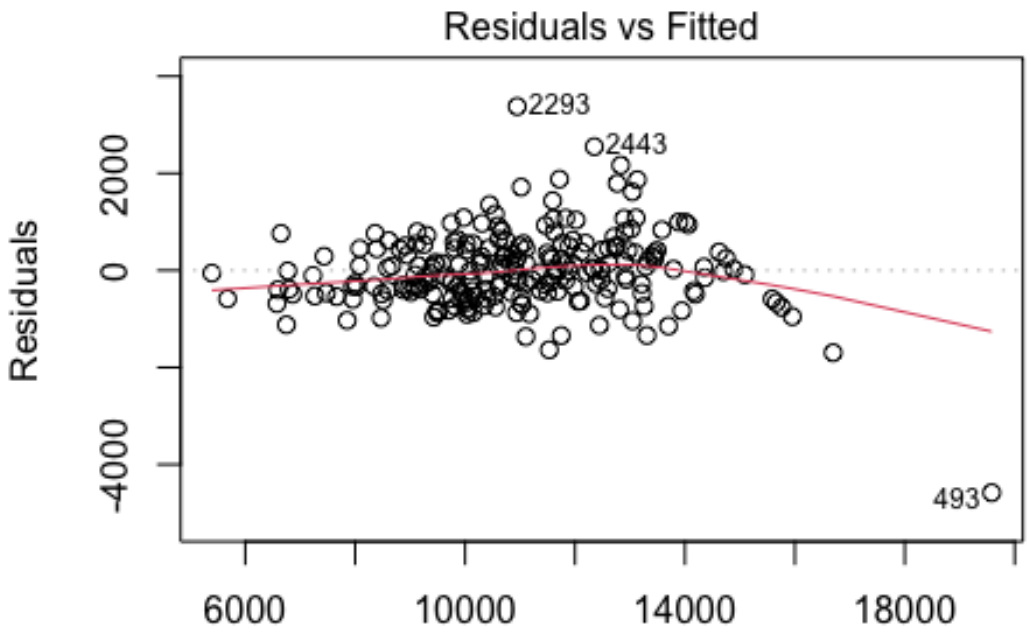
```
final_model.lm <- step(initial_model.lm)
```

```
## Start:  AIC=3160.61
## carbon_footprint ~ annual_energy_consumption + income + household_size +
##     environmental_concern_score + diet_type + uses_public_transport +
##     home_type
##
##                                Df Sum of Sq       RSS    AIC
## - diet_type                     3     93942 139732118 3154.8
## - environmental_concern_score   1    389704 140027880 3159.3
## <none>                                      139638176 3160.6
## - household_size                1   1346443 140984619 3160.9
## - home_type                     3   3878661 143516837 3161.1
## - uses_public_transport         1   1653630 141291806 3161.4
## - income                        1 239876295 379514471 3394.6
## - annual_energy_consumption     1 561858851 701497027 3539.6
##
## Step:  AIC=3154.77
## carbon_footprint ~ annual_energy_consumption + income + household_size +
##     environmental_concern_score + uses_public_transport + home_type
##
##                                Df Sum of Sq       RSS    AIC
```

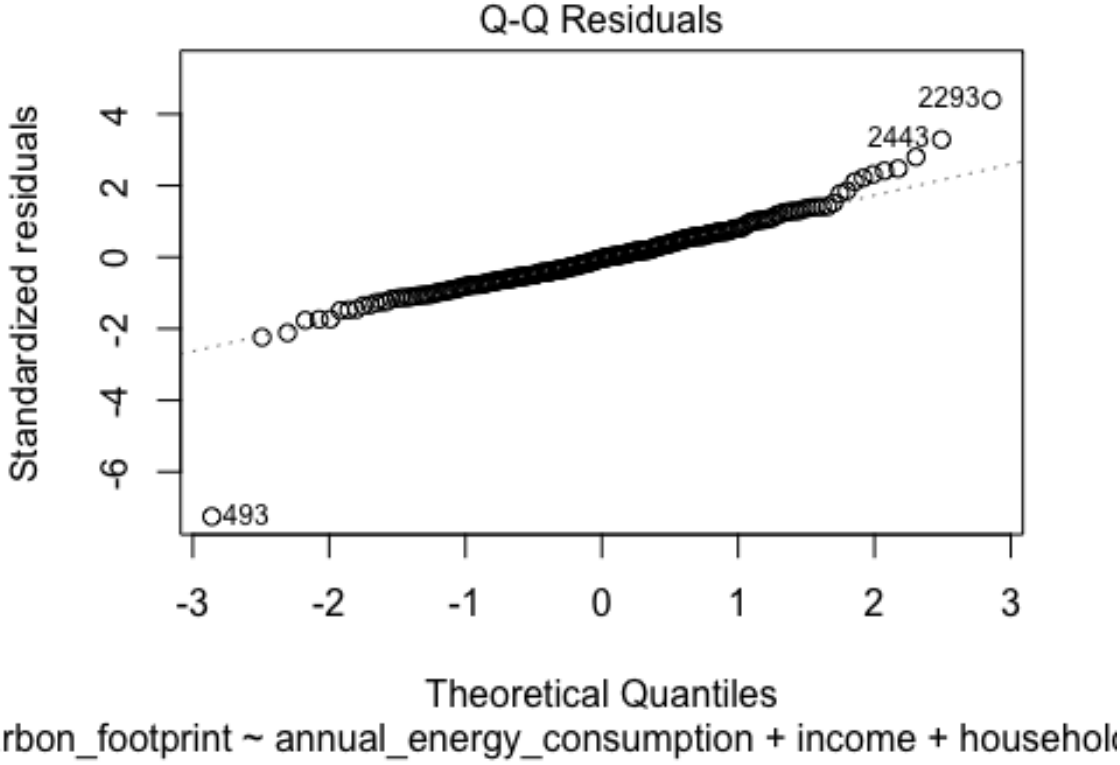
```
## - environmental_concern_score  1    383332 140115450 3153.4
## <none>                                     139732118 3154.8
## - household_size               1   1501260 141233378 3155.3
## - home_type                    3   3963067 143695185 3155.4
## - uses_public_transport        1   1647791 141379910 3155.5
## - income                       1 240438459 380170577 3389.0
## - annual_energy_consumption    1 568995205 708727323 3536.0
##
## Step:  AIC=3153.42
## carbon_footprint ~ annual_energy_consumption + income + household_size +
##     uses_public_transport + home_type
##
##                              Df Sum of Sq       RSS    AIC
## <none>                                     140115450 3153.4
## - household_size              1   1433584 141549034 3153.8
## - home_type                   3   3858597 143974047 3153.8
## - uses_public_transport       1   1465020 141580470 3153.9
## - income                      1 243412043 383527493 3389.1
## - annual_energy_consumption   1 583120527 723235977 3538.8
```

Proposed model diagnostics

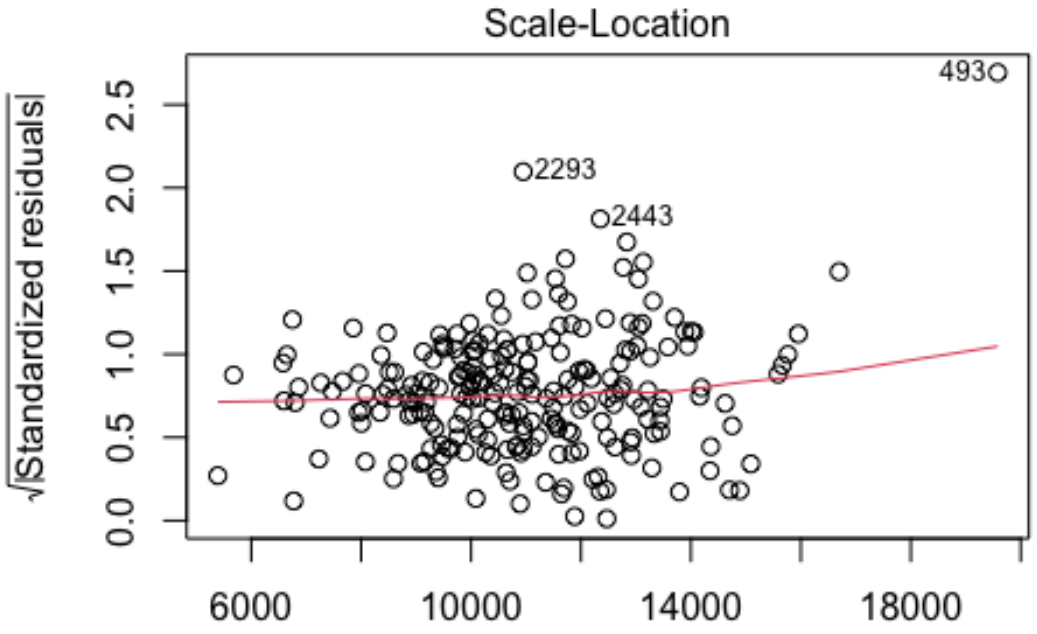
```
plot(final_model.lm)
```



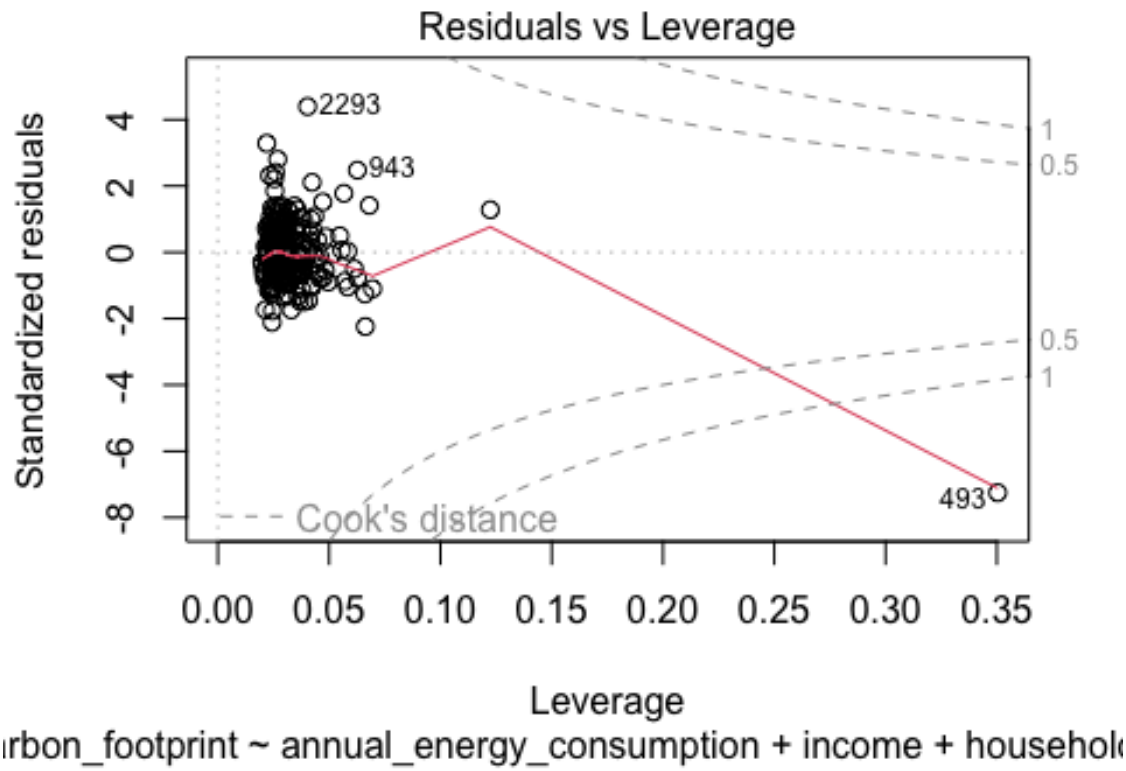
ırbon_footprint ~ annual_energy_consumption + income + household



ırbon_footprint ~ annual_energy_consumption + income + household



ırbon_footprint ~ annual_energy_consumption + income + household



ırbon_footprint ~ annual_energy_consumption + income + household

Diagnostic plots for the final model indicate mild heteroscedasticity, with residuals slightly spreading more at higher fitted values, and minor deviations from normality in the tails of the Q–Q plot. A few observations were identified, but none exceeded Cook's distance thresholds.

Proposed model intepretation

```
summary(final_model.lm)

##
## Call:
## lm(formula = carbon_footprint ~ annual_energy_consumption + income +
##     household_size + uses_public_transport + home_type, data = mydf)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4581.3  -465.0   -27.7   441.6  3377.2
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 4.269e+02  2.909e+02   1.468   0.1436
## annual_energy_consumption   1.988e+00  6.455e-02  30.804   <2e-16 ***
## income                      6.921e-02  3.478e-03  19.902   <2e-16 ***
```

```
## household_size              6.579e+01  4.308e+01   1.527   0.1281
## uses_public_transportYes   -1.580e+02  1.023e+02  -1.544   0.1240
## home_typeFlat               1.137e+02  1.472e+02   0.773   0.4405
## home_typeSemi-detached      1.414e+02  1.491e+02   0.948   0.3440
## home_typeTerraced           3.475e+02  1.425e+02   2.439   0.0155 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 783.9 on 228 degrees of freedom
## Multiple R-squared:  0.8824, Adjusted R-squared:  0.8788
## F-statistic: 244.5 on 7 and 228 DF,  p-value: < 2.2e-16
```

The proposed model shows that annual_energy_consumption is the primary driver of carbon_footprint, with higher energy use leading to substantially increased emissions. Income also has a strong positive effect, suggesting wealthier households generate larger carbon footprints. household_size and home_type introduce additional variation, whereas the variables removed during model selection showed weak or noisy patterns in the exploratory analysis.

4. Model the likelihood of being a household that regularly recycles.

The aim of this analysis is to model whether a household regularly recycles, using the binary response variable recycles_regularly. Logistic regression was used because the outcome is binary and the model estimates the probability of an event, making it appropriate for this type of problem.

Based on the exploratory data analysis, several household and behavioural variables were considered as potential explanatory variables. environment_concern_score was included as it showed a clear separation between recycling and non-recycling households and is theoretically linked to pro-environmental behaviour. diet_type was included as a behavioural indicator associated with environmental values, while home_type was retained as a household characteristic that may influence recycling; both variables showed mild distributional differences with recycling in the exploratory analysis. Additional variables such as income, household_size, annual_energy_consumption, and uses_public_transport were initially considered, but exploratory analysis suggested weak and inconsistent relationships with the recycling outcome.

A maximal logistic regression model including all candidate explanatory variables was fitted initially. Interaction terms were briefly explored, but did not improve model interpretability or fit. Stepwise model selection based on AIC was then applied to obtain a more parsimonious model that balances explanatory power and simplicity. Variables that did not improve model fit were removed during this process. The final proposed model retains environment_concern_score, diet_type, and home_type. This model showed a

substantial reduction in deviance compared to the maximal model and a lower AIC than more complex alternatives, indicating improved fit without unnecessary complexity.

The model limitation is that not all levels of the categorical predictors diet_type and home_type are significant, indicating variation in their relationships with recycling behaviour. However, these variables were retained because exploratory analysis and stepwise selection showed that, as grouped predictors, they improved model fit and added meaningful context. Overall, the final model is interpretable, grounded in theory, and suitable for explaining household recycling behaviour in this dataset.

Proposed Model

```
recycle_model.glm <- glm( recycles_regularly ~ environmental_concern_score +
diet_type + home_type, data = mydf, family = binomial)
```

Model intepretation

The model summary shows that environmental_concern_score has a strong, statistically significant positive association with the likelihood of regular recycling. Certain levels of home_type, especially flats, show positive effects, while diet_type coefficients are generally weaker, with vegan households showing a negative association relative to the reference category. The model greatly improves fit relative to the null model, as evidenced by the considerable reduction in deviance, supporting the relevance of the included predictors despite variability within categorical levels.

The odds ratios indicate that environmental_concern_score is the strongest predictor of regular recycling, with a one-unit increase multiplying the odds by more than six, demonstrating a substantial positive effect. diet_type shows mixed associations: vegan households have lower odds of recycling regularly than the reference group, while other dietary categories show weaker differences. home_type also influences recycling likelihood, with flats showing significantly higher odds of regular recycling (around 5 times), while semi-detached and terraced dwellings show more moderate increases.

```
summary(recycle_model.glm)

##
## Call:
## glm(formula = recycles_regularly ~ environmental_concern_score +
##     diet_type + home_type, family = binomial, data = mydf)
##
## Coefficients:
##                               Estimate Std. Error z value Pr(>|z|)
## (Intercept)                  -11.29649    1.62810  -6.938 3.96e-12 ***
## environmental_concern_score    1.82123    0.25165   7.237 4.58e-13 ***
## diet_typePescatarian           0.05915    0.86714   0.068  0.94562
## diet_typeVegan                -1.18041    0.58758  -2.009  0.04454 *
## diet_typeVegetarian            0.44947    0.56998   0.789  0.43036
## home_typeFlat                  1.76509    0.64149   2.752  0.00593 **
## home_typeSemi-detached         1.02897    0.61962   1.661  0.09678 .
```

```
## home_typeTerraced             1.11920    0.62631   1.787  0.07394 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 326.33  on 235  degrees of freedom
## Residual deviance: 144.27  on 228  degrees of freedom
## AIC: 160.27
##
## Number of Fisher Scoring iterations: 6
```

```
exp(coef(recycle_model.glm))
```

```
##                 (Intercept) environmental_concern_score
##               1.241638e-05                6.179433e+00
##        diet_typePescatarian              diet_typeVegan
##               1.060930e+00                3.071524e-01
##         diet_typeVegetarian                home_typeFlat
##               1.567485e+00                5.842126e+00
##      home_typeSemi-detached           home_typeTerraced
##               2.798190e+00                3.062394e+00
```